Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In a hydrogen atom following the Bohr postulates the product of linear momentum and angular momentum is proportional to $(n)^x$ where 'n' is the orbit number. Then 'x'is-

(A) 0

(B) 2

(C) -2

(D) 1

Q.2

In the hydrogen atom, an electron makes a transition from $n = 2$ to $n = 1$. The magnetic field produced by the circulating electron at the nucleus -

(A) decreases 16 times

(B) increases 4 times

(C) decreases 4 times

(D) increases 32 times

Q.3

Which of the following statements are true regarding Bohr's model of hydrogen atom?
(I) Orbiting speed of electron decreases as it shifts to discrete orbits away from the nucleus
(II) Radii of allowed orbits of electron are proportional to the principal quantum number
(III) Frequency with which electrons orbit around the nucleus in discrete orbits is inversely proportional to the cube of principal quantum number
(IV) Binding force with which the electron is bound to the nucleus increases as it shifts to outer orbits
Select correct answer using the codes given below.
Codes:

(A) I and III

(B) II and IV

(C) I, II and III

(D) II, III and IV

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Q.4

Suppose an electron is attracted towards the origin by a force $\frac{k}{r}$ where k is a constant and r is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the nth orbital of the electorn is found to be 'r' and the kinetic energy of the electron to be ' K_n ' Then which of the following is true

(A) K_n independent of $n, r_n \propto n$

(B) $K_n \propto \frac{1}{n}, r_n \propto n$

(C) $K_n \propto \frac{1}{n}, r_n \propto n^2$

(D) $K_n \propto \frac{1}{n^2}, r_n \propto n^2$

Q.5

An electron in a hydrogen atom makes a transition from $n = n_1$ to $n = n_2$. The time period of electron in the initial state is eight times that in the final state. Then which of the following statement is true ?

(A) $n_1 = 3n_2$

(B) $n_1 = 4n_2$

(C) $n_1 = 2n_2$

(D) $n_1 = 5n_2$

Q.6

If the angular momentum of an electron in an orbit is J then the K.E. of the electron in that orbit is

(A) $\frac{\mathrm{J}^2}{2\mathrm{mr}^2}$

(B) $\frac{\mathrm{J}^2}{2\ \mathrm{m}}$

(C) $\frac{\mathrm{J}^2}{2\pi}$

(D) $\frac{\mathrm{Jv}}{\mathrm{r}}$

Q.7

A neutron travelling with a velocity v and kinetic energy E has a perfectly elastic head-on collision with a nucleus of an atom of mass number A at rest. The fraction of total energy retained by the neutron is approximately

(A) $[(A-1)/(A+1)]^2$

(B) $[(A+1)/(A-1)]^2$

(C) $[(A-1)/A]^2$

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(D) $[(A+1)/A]^2$

Q.8

The energy difference between the first two levels of hydrogen atom is $10.2\mathrm{eV}$ for another element of atomic number 10 and mass number 20, this will be

(A) $2040\mathrm{eV}$

(B) $0.201\mathrm{eV}$

(C) $510\mathrm{eV}$

(D) $1020\mathrm{eV}$

Q.9

If a proton had a radius R and the charge was uniformly distributed, calculate using Bohr theory, the total ground state energy of a H -atom when $\mathrm{R}=0.1 \mathring{A}$.

(A) $-13.6\mathrm{eV}$

(B) $-5.6\mathrm{eV}$

(C) $-3.67\mathrm{eV}$

(D) $-2.67\mathrm{eV}$

Q.10

An unknown hot gas emits radiation of wavelengths $46.0\ \mathrm{nm}, 82.8\ \mathrm{nm}$ and $103.5\ \mathrm{nm}$ only. Assume that the atoms have only two excited states and the difference between consecutive energy levels decreases as energy is increased. Taking the energy of the highest energy state to be zero, find the energies of the ground state and the first excited state.

(A) $-27\mathrm{eV}, -12\mathrm{eV}$

(B) $-6\mathrm{eV}, -3\mathrm{eV}$

(C) $-11\mathrm{eV}, -8\mathrm{eV}$

(D) $-9\mathrm{eV}, -3\mathrm{eV}$

Q.11

Consider 3rd orbit of He^{+}(Helium), using nonrelativistic approach, the speed of electron in this orbit will be [given $\mathrm{K}=9 \times 10^9$ constant, $\mathrm{Z}=2$ and h (Plank's Constant) $=6.6 \times 10^{-34} \mathrm{Js}$]

(A) $1.46 \times 10^6\ \mathrm{m/s}$

(B) $0.73 \times 10^6\ \mathrm{m/s}$

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(C) $3.0 \times 10^8 \ \mathrm{m/s}$

(D) $2.92 \times 10^6 \ \mathrm{m/s}$

Q.12

A hydrogen atom in state $\mathrm{n} = 6$ makes two successive transitions and reaches the ground state. In the first transition a photon of $1.13\mathrm{eV}$ is emitted. Find the energy of the photon emitted in the second transition and the value of n in the intermediate state.

(A) $12.1\mathrm{eV}, \mathrm{n} = 3$

(B) $6.1\mathrm{eV}, \mathrm{n} = 4$

(C) $2.1\mathrm{eV}, \mathrm{n} = 3$

(D) $1.1\mathrm{eV}, \mathrm{n} = 5$

Q.13

Electrons in a certain energy level $\mathrm{n} = \mathrm{n}_1$, can emit 3 spectral lines. When they are in another energy level, $\mathrm{n} = \mathrm{n}_2$. They can emit 6 spectral lines. The orbital speed of the electrons in the two orbits are in the ratio of

(A) $4 : 3$

(B) $3 : 4$

(C) $2 : 1$

(D) $1 : 2$

Q.14

The wavelength of K_a X-rays produced by an X-ray tube is $0.76\mathring{A}$. Find the atomic number of the anode material of the tube.

(A) 40

(B) 30

(C) 20

(D) 10

Q.15

A hydrogen atom in its ground state absorbs $10.2\mathrm{eV}$ of energy. The orbital angular momentum is increased by

(A) $1.05 \times 10^{-34} \ \mathrm{J-s}$

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(B) 3.16×10^{-34} J $-$ s

(C) 2.11×10^{-34} J $-$ s

(D) 4.22×10^{-34} J $-$ s

Q.16

Electrons are bombarded to excite hydrogen atoms and six spectral lines are observed. If Eg is the ground state energy of hydrogen, the minimum energy the bombarding electrons should posses is

(A) $\frac{8E_g}{9}$

(B) $\frac{15E_g}{16}$

(C) $\frac{35E_g}{36}$

(D) $\frac{48E_g}{49}$

Q.17

If the atom ${}_{100}Fm^{257}$ follows the Bohr model and the radius of ${}_{100}Fm^{257}$ is n times the Bohr radius, then find n.

(A) 100

(B) 200

(C) 4

(D) $1/4$

Q.18

In Bohr theory of hydrogen atom, let r, v and E be the radius of orbit, speed of electron and the total energy of the electron respectively. Which of the following quantities is proportional to the quantum number n ?

(A) vr

(B) rE

(C) r/E

(D) r/v

Q.19

The ionization energy of hydrogen-like Bohr atom is 4 Rydbergs. Find the wavelength of radiation emitted when the electron jumps from the first excited state to the ground state:
[1 Rydberg $= 2.2 \times 10^{-18}, h = 6.6 \times 10^{-34}$Js,

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$c = 3 \times 10^8\ \mathrm{m/s}$
Bohr radius of hydrogen atom $= 5 \times 10^{-11}\ \mathrm{m}]$

(A) $400\mathring{A}$

(B) $300\mathring{A}$

(C) $500\mathring{A}$

(D) $600\mathring{A}$

Q.20

Suppose potential energy between electron and proton at separation r is given by $U = \mathrm{K}\ln(r)$, where K is a constant. For such a hypothetical hydrogen atom, the ratio of energy difference between energy levels $(n = 1$ and $n = 2)$ and $(n = 2$ and $n = 4)$ is

(A) 1

(B) 2

(C) 3

(D) 4

Answers & Solutions

Q.1 Answer:

0

Solution:

Linear momentum $\Rightarrow \mathrm{mv}\ \alpha 1/\mathrm{n}$
angular momentum $\Rightarrow$ **mv** αn
$\therefore$ **product of linear momentum and angular momentum** αn^0

Q.2 Answer:

increases 32 times

Solution:

$\because \quad \mathrm{B} = \frac{\mu_0 \mathrm{I}}{2\mathrm{r}}$ **and** $\mathrm{I} = \frac{\mathrm{e}}{\mathrm{T}}$
$\mathrm{B} = \frac{\mu_0 \mathrm{e}}{2\mathrm{rT}} \quad \left[\mathrm{r} \propto \mathrm{n}^2,\ \mathrm{T} \propto \mathrm{n}^5\right]$
$\mathrm{B} \propto \frac{1}{\mathrm{n}^5}$

Q.3 Answer:

I and III

Solution:

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Orbital speed varies inversely as the radius of the orbit.

$\mathrm{v} \propto \frac{1}{\mathrm{n}}$

Q.4 Answer:

K_n independent of $n, r_n \propto n$

Solution:

$\frac{mv^2}{r} = \frac{K}{r}$ **or** $v = \sqrt{\frac{K}{m}} = cr^\circ$

Now $mvr = \frac{nh}{2\pi}$ **or** $r = \frac{nh}{2\pi mv}$

$\therefore r \cdot \propto n.$

Also kinetic energy $K = \frac{mv^2}{2} = \frac{m}{2} \times \frac{K}{m} = \frac{K}{2}.$

Q.5 Answer:

$n_1 = 2n_2$

Solution:

In the nth orbit, let r_n **be the radius and** v_n **be the speed of electron. Time period,** $T_n = \frac{2\pi r_n}{v_n} \propto \frac{r_n}{v_n}$

Now $r_n \propto n^2; v_n \propto \frac{1}{n} \therefore \frac{r_n}{v_n} \propto n^3$ **or** $T_n \propto n^3$

Here $8 = \left(\frac{n_1}{n_2}\right)^3$ **or** $\frac{n_1}{n_2} = 2$ **i.e.,** $n_1 = 2n_2$

Q.6 Answer:

$\frac{\mathrm{J}^2}{2\mathrm{mr}^2}$

Solution:

Angular momentum = mrv = J

$\therefore \mathrm{V} = \frac{J}{\mathrm{mr}}$

K. E. of electron $= \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{J}{mr}\right)^2 = \frac{J^2}{2mr^2}$

Q.7 Answer:

$[(A-1)/(A+1)]^2$

Solution:

$v_1' = \frac{(m_1 - m_2)v_1 + 2m_2v_2}{m_1 + m_2}$

As v_2 **is zero,** $m_2 > m_1, v_1'$ **is in the opposite direction.** $m_1 = 1, m_2 = A. \quad \therefore |v_1'| = \frac{(A-1)}{(A+1)}v_1$

The fraction of total energy retained is

$\frac{1/2mv_1'^2}{1/2v_1^2} = \frac{(A-1)^2}{(A+1)^2}$

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Q.8 Answer:

1020eV

Solution:

For H atom

$E_2 - E_1 = 10.2\text{eV}$

For $Z = 10, E_1 = \frac{-13.6\times10^2}{1^2} = -1360\text{eV}$

$E_2 = \frac{-13.6\times10^2}{2^2} = -3.4 \times 100 = -340\text{eV}$

$\therefore \quad E_2 - E_1 = -340 - (-1360) = 1020\text{eV}$

Q.9 Answer:

-13.6eV

Solution:

The electron present in H**-atom revolves around the point size proton of a defined radius** rB **(Bohr's radius) in a circular path when the atom is in the ground state.**

$As\ \text{mvr}_B = h$ **and** $\frac{Mv^2}{r_B} = \frac{-1\times e\times e}{4\pi\varepsilon_0 r_B^2}$

$$\frac{m}{r_B}\left(\frac{-h^2}{m^2 r_B^2}\right) = \frac{e^2}{4\pi\epsilon_0 r_B^2}$$

$$r_B = \frac{4\pi\epsilon_0}{e^2}\frac{h^2}{m} = 0.53\dot{A}$$

$$K.E = \frac{1}{2}mv^2 = \left(\frac{m}{2}\right)\left(\frac{h}{mr_B}\right)^2$$

$$= \frac{h^2}{2mr_B^2} = 13.6\text{eV}$$

total enrgy of the electron i.e

$$\text{E} = \text{K} + \text{U} = +13.6\text{eV} - 27.2\text{eV} = -13.6\text{eV}$$

when $r = 0.1\mathring{A} : R < r_B$ **(as** $r_B = 0.51\mathring{A}$ **) and the ground state energy is the same as obtained earlier for point size proton i.e** -13.6eV

Q.10 Answer:

$-27\text{eV}, -12\text{eV}$

Solution:

Energy in ground state is the energy acquired in the transition of second excited state to ground state.

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As second excited state is taken as zero level.

$$E = \frac{hc}{\lambda_1}$$
$$= \frac{4.14*10^{-15}*3*10^8}{46*10^{-9}} = \frac{1242}{46} = 27\text{eV}$$

Again energy in the first excited state

$$E = \frac{hc}{\lambda_{11}}$$
$$= \frac{4.14*10^{-15}*3*10^8}{103.5} = 12\text{eV}$$

Q.11 Answer:

$1.46 \times 10^6\ \text{m/s}$

Solution:

Speed of electron in nth **orbit** $\text{V}_\text{n} = \frac{2\pi\text{KZe}^2}{\text{nh}}$

$\text{V} = \left(2.19 \times 10^6\ \text{m/s}\right)\frac{\text{Z}}{\text{n}}$

$\text{V} = \left(2.19 \times 10^6\right)\frac{2}{3}(\text{Z} = 2\&\text{n} = 3)$

$\text{V} = 1.46 \times 10^6\ \text{m/s}$

Q.12 Answer:

$12.1\text{eV}, \text{n} = 3$

Solution:

Given: $n = 6$ **to** $n = 1$.

Let after first transition hydrogen atom reaches n^{th} **state.**

Emergy released in first transition $= 1.13\text{ev}$.

$$= 1.13\text{ev}$$

Now we have to energy released in second transition,

$$E_n = -13.6 - [1.13 + (-0.377)]$$
$$E_n = -12.1\text{ev}$$

Hence 12.1ev is released in second transition.

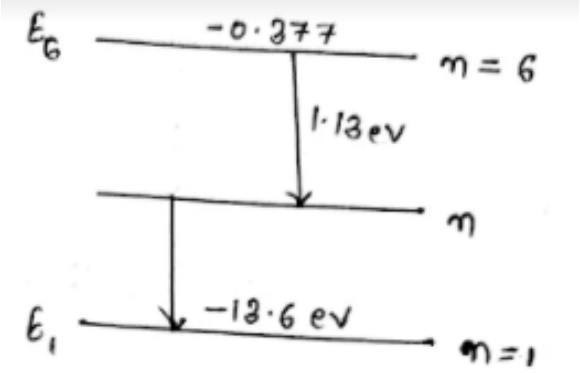
Now, we have to find intermediate state i.e n.

$\because$ **Energy of** n^{th} **state is** $[-13.6 - (12.1)] = -1.5\text{ev}$.

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Now we know, $E_\eta = -\frac{13.6}{n^2}z^2$

$$E_n = -1.5\text{ev (here)}$$
$$z = 1$$
$$\because \quad -1.5 = -\frac{13.6}{n^2}$$
$$n^2 \approx 9$$
$$n = 3$$



Q.13 Answer:

$4 : 3$

Solution:

Number of emission spectral lines, $\mathrm{N} = \frac{\mathrm{n}(\mathrm{n}-1)}{2} \therefore 3 = \frac{\mathrm{n}_1(\mathrm{n}_1-1)}{2}$**, in first case.**

$$\mathrm{n}_1^2 - \mathrm{n}_1 - 6 = 0 \text{ or } (\mathrm{n}_1 - 3)(\mathrm{n}_1 + 2) = 0$$

Take positive root.

$$\therefore \mathrm{n}_1 = 3$$

Again, $6 = \frac{\mathrm{n}_2(\mathrm{n}_2-1)}{2}$**, in second case.**

$$\mathrm{n}_2^2 - \mathrm{n}_2 - 12 = 0 \text{ or } (\mathrm{n}_2 - 4)(\mathrm{n}_2 + 3) = 0$$

Take positive root, or $n_2 = 4$

Now velocity of electron $\mathrm{v} = \frac{2\pi \mathrm{KZe}^2}{\mathrm{nh}} \therefore \frac{\mathrm{v}_1}{\mathrm{v}_2} = \frac{\mathrm{n}_2}{\mathrm{n}_1} = \frac{4}{3}$

Q.14 Answer:

40

Solution:

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$\frac{1}{\lambda} = RZ^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3}{4}\mathrm{RZ}^2$
or $Z^2 = \frac{4}{3R\lambda}$
$= \frac{4}{3\mathrm{x}(1.097\times10^7)\mathrm{x}(0.76\times10^{-10})} = 1599.25$
or $Z^2 = 1600 \quad \therefore \quad Z = 40.$

Q.15 Answer:

$1.05 \times 10^{-34}\ \mathrm{J-s}$

Solution:

Electron after absorbing $10.2\mathrm{eV}$ energy goes to its first excited state $(\mathrm{n}=2)$ from ground state $(\mathrm{n}=1)$.
$\therefore$ Increase in momentum $= \frac{\mathrm{h}}{2\pi}$
$= \frac{6.6\times10^{-34}}{6.28} = 1.05 \times 10^{-34}\ \mathrm{J-s}$

Q.16 Answer:

$\frac{15E_g}{16}$

Solution:

To obtain 6 spectral line, as electron must be excited to fourth orbit with energy $\frac{E_g}{16}$, so that the difference is $\frac{15E_g}{16}$.

Q.17 Answer:

$1/4$

Solution:

Bohr's radius is given for n^{th} orbit is given by $r_m = \frac{I_0 m^2}{Z}$ where r_0 is bohr's radius and m is orbit number.
For $\mathrm{F_m}, \mathrm{m}=5$ (Fifth orbit in which the outermost electron is present)

$$\mathrm{r_m} = \frac{\mathrm{r_0 \cdot 5^2}}{100} = \mathrm{nr_o}$$

$$\mathrm{n} = 1/4$$

Q.18 Answer:

vr

Solution:

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Note that v is inversely proportional to n whereas r is directly to n^2

Q.19 Answer:

$300\mathring{A}$

Solution:

The energy in the ground state

$E_1 = -4$ **Rydberg** $\quad E_1 = -4 \times 2.2 \times 10^{-18}$ J

The energy of the first excited state $(n = 2)\ E_2 = \frac{E_1}{4} = -2.2 \times 10^{-18} J$

The energy difference $\Delta E = E_2 - E_1 = 3 \times 2.2 \times 10^{-18}$ J

Now, the wavelength of radiation emitted is $\lambda = \frac{hc}{\Delta E}$

$\lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{3 \times 2.2 \times 10^{-18}} = 300\mathring{A}$

Q.20 Answer:

1

Solution:

$-\frac{dU}{dr} = F$ **(conservative force field)**

$\Rightarrow \quad F = \frac{-K}{r}$ **provides the centrifugal force for circular motion of electron.** $\frac{mv^2}{r} = \frac{\mathrm{K}}{r} \Rightarrow r = \frac{nh}{2\pi\sqrt{m\,\mathrm{K}}}$

K.E. of electron $= \frac{1}{2}mv^2 = \frac{1}{2}\,\mathrm{K}$

P.E. of electron $= \mathrm{K} \ln r\ E(n) =$ **Total energy = K.E. + P.E.** $= \frac{1}{2}\,\mathrm{K} + \mathrm{K} \ln r = \frac{\mathrm{K}}{2}\left[1 + \log \frac{n^2h^2}{4\pi^3 mk}\right]$

Required ratio $= \frac{E(2) - E(1)}{E(4) - E(2)} = 1$

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